Aquablend — MILP Mathematical Formulation

MILP Stream

1 Overview

Water corporations draw drinking water from multiple sources, such as reservoirs, river intakes, and groundwater bores, that differ in cost and in raw water quality (pH, alkalinity, turbidity, and so on). Before that water can be supplied for human consumption, it is treated at a plant and delivered to demand zones. The question this model answers is: which sources should be drawn from, how much from each, and how should that water be routed to plants and zones, so that demand is met, and water quality stays within safe limits, at the lowest possible cost?

This is formulated as a Mixed-Integer Linear Program (MILP). Whether a source, plant, or transfer link is used at all is a binary decision; how much water is drawn or flows along each link is a continuous decision. The objective balances fixed activation costs against the variable cost of drawing, transferring and treating water.

Treatment is modelled in two parts. A flat rate per megalitre (ML) covers the cost of operating a plant, while each removal process is charged based on the load it removes. The load-based removal cost makes source blending an economic decision rather than purely a feasibility problem: a cheaper but more contaminated source may incur higher treatment costs than a cleaner, more expensive source. Treatment here can only reduce a parameter, never add to it. Chemical dosing, which allows for a lower limit to be met by treatment rather than by source selection, is a future enhancement.

The toy case considers the simplest version of this problem: one treatment plant and one demand zone, fed by several candidate sources. With only one plant and one zone, there is no routing choice to make: every active source sends its full withdrawal to the plant, and the plant sends its entire outflow to the demand zone. As a result, the routing variables are determined by the source withdrawals, leaving the model to determine which sources to activate, how much to draw from each, and any required treatment decisions. Section 7 derives this toy case as a special case of the general network formulation, so the same formulation can later be extended to multiple plants and zones without being rebuilt from scratch.

2 Notation

Convention. This is a single-period model (one representative day), so no day index is used. Parameters are uppercase; decision variables are lowercase, using Greek letters for binary decisions and lowercase Latin for continuous quantities. A subscript is an index; an overbar/underbar is an upper/lower bound.

2.1 Sets

Table 1: Notations on sets of input parameters

Notation	Description
$\mathcal{S}$	set of water sources, $s \in \mathcal{S}$
$\mathcal{T}$	set of treatment plants, $t \in \mathcal{T}$
$\mathcal{Z}$	set of demand zones, $z \in \mathcal{Z}$
$\mathcal{P}$	set of water quality parameters, $p \in \mathcal{P}$
$\mathcal{R}$	set of removal processes, $r \in \mathcal{R}$

Continued on next page

Notation	Description
$\mathcal{R}_t$	processes installed at plant t, as an ordered sequence $r_1, \dots, r_{n_t}$ drawn from $\mathcal{R}$, in the order water flows through them

$n_t = |\mathcal{R}_t|$ is written with the index because plants differ in how many processes they run. For example, a plant with coagulation, filtration and chlorination has $n_t = 3$, while one with filtration alone has $n_t = 1$.

2.2 Parameters

Table 2: Notations on input parameters

Notation	Description	Units
Demand		
D_z	volume of water to deliver to demand zone z	ML/day
Sources		
F_s	fixed cost of activating source s	\$/day
C_s	unit cost of drawing water from source s	\$/ML
$\underline{W}_s$	minimum daily withdrawal from source s	ML/day
$\overline{W}_s$	maximum daily withdrawal from source s	ML/day
Treatment plants		
F_t	fixed cost of activating treatment plant t	\$/day
C_t	unit treatment cost at plant t	\$/ML
$\underline{V}_t$	minimum daily throughput of plant t	ML/day
$\overline{V}_t$	maximum daily throughput of plant t	ML/day
Arcs		
$\overline{L}_{st}$	maximum flow on the arc from source s to plant t	ML/day
$\overline{L}_{tz}$	maximum flow on the arc from plant t to zone z	ML/day
C_{st}	unit cost of transferring water from source s to plant t	\$/ML
C_{tz}	unit cost of transferring water from plant t to zone z	\$/ML
Water quality		
Q_{sp}	value of quality parameter p in source s	units of p
$\underline{Q}_p$	lower regulatory limit for parameter p	units of p
$\overline{Q}_p$	upper regulatory limit for parameter p	units of p
$\underline{I}_{tp}$	minimum value of p that plant t can accept	units of p
$\overline{I}_{tp}$	maximum value of p that plant t can accept	units of p
Removal processes		
F_{tr}	fixed daily cost of operating process r at plant t	\$/day
C_{trp}	cost per unit of load of p removed by process r at plant t	\$/(units of p·ML)
H_{trp}	maximum fraction of the load of p removable by process r at plant t, with $0 \le H_{trp} \le 1$	—

Costs and limits are treated as fixed. The units of a quality parameter depend on p (e.g. mg/L, NTU). Quality parameters that are not approximately linear in their natural units are represented using transformed quantities chosen to support linear blending within the MILP.

The inlet limits $\underline{I}_{tp}$ and $\overline{I}_{tp}$ bound the same physical quantity as the regulatory limits $\underline{Q}_p$ and $\overline{Q}_p$, but they have different meanings. I is what a plant can accept, while Q denotes the standards that the water must meet after treatment before reaching the demand zone.

C_{trp} is charged per unit of load removed. When p is represented as a concentration, multiplying concentration by volume gives a quantity proportional to mass, subject to consistent units. For parameters such as turbidity measured in NTU, the cost is per NTU·ML, which is not a mass; however, we assume it correlates to the mass of sediment that would produce that level of turbidity.

2.3 Decision variables

Just for notation consistency. Since in **Tab. 2**, you used C_{st} to represent cost from source s to plant t. Should we also use c_{st} instead of b_{st}?

Table 3: Notations on decision variables

Notation	Domain	Description	Units
Sources			
α_s	$\{0, 1\}$	1 if source s is activated	—
a_s	≥ 0	volume drawn from source s	ML/day
Treatment plants			
β_t	$\{0, 1\}$	1 if plant t is activated	—
Arcs			
γ_{st}	$\{0, 1\}$	1 if source s sends water to plant t	—
b_{st}	≥ 0	volume of water from source s to plant t	ML/day
δ_{tz}	$\{0, 1\}$	1 if plant t sends water to demand zone z	—
c_{tz}	≥ 0	volume of water from plant t to demand zone z	ML/day
Removal processes			
σ_{tr}	$\{0, 1\}$	1 if removal process r is operated at plant t	—
e_{trp}	≥ 0	load of parameter p removed by process r at plant t	(units of p)·ML/day
ℓ^i_{tp}	≥ 0	load of p remaining after the i-th process at plant t	(units of p)·ML/day

σ_{tr} and e_{trp} are defined only for $r \in \mathcal{R}_t$, and ℓ^i_{tp} only for $i = 0, \ldots, n_t$; the objective and constraints below reflect this. Since we have $\mathcal{R}_t = (r_1, \ldots, r_{n_t})$ (Table 1), the process at stage i is r_i: variables are indexed by the process itself, so e_{tr_ip} is the same variable as e_{trp} with $r = r_i$.

3 Example network structure

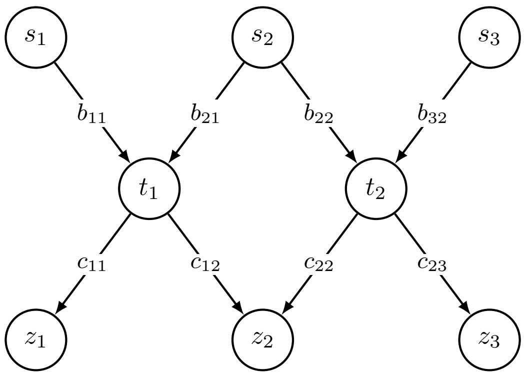

Figure 1: One example of the network structure (sources → treatment plants → demand zones). Each source-plant arc has an activation binary γ_{st} and a flow variable b_{st}; each plant-zone arc has δ_{tz} and c_{tz}. The formulation is generic over any number of sources, plants, zones and links.

4 Objective

Minimise total cost over a single period.

Costs are grouped according to the part of the network to which they belong: sources, arcs, plants, and removal processes. Sources and plants carry fixed activation costs and variable costs based on the volume handled. Removal processes carry a fixed operating cost and a variable cost based on the load removed. Arcs are charged on flow volume only, since the model does not currently assign a fixed cost to activating a link.

The removal cost makes source blending an economic decision rather than purely a feasibility problem. It is the only cost term that depends directly on water quality; all other variable costs depend only on flow volume. The coefficient C_{trp} is charged per unit of parameter p removed by process r at plant t, so a source requiring less removal may incur lower treatment costs.

$$
\begin{aligned}
\min \quad & \sum_{s\in\mathcal{S}} F_s \alpha_s && \text{source fixed cost} \\
+ & \sum_{s\in\mathcal{S}} C_s a_s && \text{source draw cost} \\
+ & \sum_{t\in\mathcal{T}} F_t \beta_t && \text{plant fixed cost} \\
+ & \sum_{t\in\mathcal{T}} C_t \sum_{s\in\mathcal{S}} b_{st} && \text{plant throughput cost} \\
+ & \sum_{s\in\mathcal{S}} \sum_{t\in\mathcal{T}} C_{st}\, b_{st} + \sum_{t\in\mathcal{T}} \sum_{z\in\mathcal{Z}} C_{tz}\, c_{tz} && \text{arc transfer cost} \\
+ & \sum_{t\in\mathcal{T}} \sum_{r\in\mathcal{R}_t} F_{tr} \sigma_{tr} && \text{removal process fixed cost} \\
+ & \sum_{t\in\mathcal{T}} \sum_{r\in\mathcal{R}_t} \sum_{p\in\mathcal{P}} C_{trp}\, e_{trp} && \text{removal cost}
\end{aligned}
\tag{1}
$$

5 Constraints

Non-negativity. All continuous variables must be non-negative.

$$
a_s,\ b_{st},\ c_{tz},\ e_{trp},\ \ell^i_{tp} \ \geq\ 0, \qquad \forall s\in\mathcal{S},\, t\in\mathcal{T},\, z\in\mathcal{Z},\, r\in\mathcal{R}_t,\, p\in\mathcal{P},\, i=0,\dots,n_t. \tag{2}
$$

Demand satisfaction. The total volume of water sent from all treatment plants to demand zone z via the arcs c_{tz} must be at least the demand D_z for that zone.

$$
\sum_{t\in\mathcal{T}} c_{tz} \geq D_z, \qquad \forall z\in\mathcal{Z}. \tag{3}
$$

Source capacity and activation. Water can only be drawn from a source if it has been activated. If source s is inactive ($\alpha_s = 0$), the draw a_s is forced to zero. If source s is active ($\alpha_s = 1$), the draw must lie between its minimum daily withdrawal $\underline{W}_s$ and maximum daily withdrawal $\overline{W}_s$.

$$
\underline{W}_s \alpha_s \leq a_s \leq \overline{W}_s \alpha_s, \quad \forall s\in\mathcal{S}. \tag{4}
$$

Plant capacity and activation. A plant can process water only if it has been activated. If plant t is inactive ($\beta_t = 0$), the total volume arriving from all sources is forced to zero. If plant t is active ($\beta_t = 1$), the total incoming volume must lie between its minimum daily throughput $\underline{V}_t$ and maximum daily throughput $\overline{V}_t$.

$$
\underline{V}_t \beta_t \leq \sum_{s\in\mathcal{S}} b_{st} \leq \overline{V}_t \beta_t, \quad \forall t\in\mathcal{T}. \tag{5}
$$

Plant flow conservation. The total volume leaving plant t toward all zones via the arcs c_{tz} equals the total volume entering it from all sources via the arcs b_{st}.

$$\sum_{z \in \mathcal{Z}} c_{tz} = \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}. \tag{6}$$

Source flow conservation. The volume drawn from each source a_s must equal the total volume it sends to all plants via the arcs b_{st}.

$$a_s = \sum_{t \in \mathcal{T}} b_{st}, \quad \forall s \in \mathcal{S}. \tag{7}$$

Arc capacity. Each arc has a maximum flow and can carry water only if that arc is active. For a source-to-plant arc, the flow b_{st} is capped at $\overline{L}_{st}$ and forced to zero unless the arc is active ($\gamma_{st} = 1$).

$$b_{st} \leq \overline{L}_{st} \gamma_{st}, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \tag{8}$$

Similarly, a plant-to-zone arc's flow c_{tz} is capped at $\overline{L}_{tz}$ and is zero unless the arc is active ($\delta_{tz} = 1$).

$$c_{tz} \leq \overline{L}_{tz} \delta_{tz}, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \tag{9}$$

Arcs require active nodes. Arcs can only be active if the corresponding upstream node (source or treatment plant) is active. This is enforced by ensuring that the binary variables representing the arcs are less than or equal to the binary variables representing the activation of the sources and plants. A source-to-plant arc requires the source to be active:

$$\gamma_{st} \leq \alpha_s, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \tag{10}$$

A plant-to-zone arc requires the plant to be active:

$$\delta_{tz} \leq \beta_t, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \tag{11}$$

Optional: a source-to-plant arc could also require the plant to be active,

$$\gamma_{st} \leq \beta_t, \quad \forall s \in \mathcal{S}, \, t \in \mathcal{T}. \tag{12}$$

This is not explicitly needed, since zero flow to an inactive plant is already guaranteed by the plant capacity constraint (5). Suppose the plant t is inactive, so $\beta_t = 0$. Then (5) becomes

$$\sum_{s \in \mathcal{S}} b_{st} \leq \overline{V}_t \, \beta_t = 0,$$

and since every flow is non-negative ($b_{st} \geq 0$), this forces $b_{st} = 0$ for all $s \in \mathcal{S}$.

Inlet treatability. A plant can only accept raw water that its installed processes are built to handle. This is a property of the plant rather than a regulatory standard, which is the reason the bound includes the index t.

$$\underline{I}_{tp} \sum_{s \in \mathcal{S}} b_{st} \leq \sum_{s \in \mathcal{S}} Q_{sp} b_{st} \leq \overline{I}_{tp} \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}, \ \forall p \in \mathcal{P}. \tag{13}$$

Each Q_{sp} must be a quantity that blends linearly by volume, so that $\sum_{s \in \mathcal{S}} Q_{sp} b_{st}$ represents the total load of parameter p arriving at the plant. The corresponding blended value is obtained by dividing this load by the total incoming water volume. $\underline{I}_{tp}$ and $\overline{I}_{tp}$ are held in the same units as Q_{sp}, so where Q_{sp} stores a transformed value the limits are transformed with it.

Not all plants impose limits on every parameter. When a plant t has no meaningful limit on the incoming p, the pair is neutralised rather than removed by setting

$$\underline{I}_{tp} = 0 \qquad \text{and} \qquad \overline{I}_{tp} = \max_{s \in \mathcal{S}} \{Q_{sp} : \overline{L}_{st} > 0\}, \tag{14}$$

where $\overline{I}_{tp}$ represents the maximum value of p among the sources that can reach t. Since the source values are non-negative, a volume-weighted blend cannot fall below zero or exceed the largest value among its inputs, so neither bound can be violated.

Removal process load balance. Water passes through the plant's removal processes sequentially, so each process acts on the load remaining after the preceding process. Denoting the load of p that remains after the i-th process as ℓ_{tp}^{i}, the load that enters the first process is what comes from the sources:

$$\ell_{tp}^{0} = \sum_{s \in \mathcal{S}} Q_{sp}\, b_{st}, \qquad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P}. \tag{15}$$

Each process r_i removes some of what it receives and passes on the rest:

$$\ell_{tp}^{i} = \ell_{tp}^{i-1} - e_{tr_i p}, \qquad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P},\ i = 1, \ldots, n_t. \tag{16}$$

The load leaving the plant is what remains after the final process, $\ell_{tp}^{n_t}$.

Removal capability. Each process removes no more than a fixed fraction $H_{tr_i p}$ of the arriving load, based on what was passed on by the previous process rather than the total load that arrived at the plant. Modelling a removal process as a fixed fraction of the incoming load is inspired by the wastewater treatment network literature [4, 3].

$$e_{tr_i p} \le H_{tr_i p}\, \ell_{tp}^{i-1}, \qquad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P},\ i = 1, \ldots, n_t. \tag{17}$$

Setting $H_{trp} = 0$ expresses that process r does not act on parameter p. By (2) and (16), no combination of processes can remove more than the load that arrived, and the series behaviour is enforced process by process.

Removal activation. No removal occurs unless the removal process is active. The coefficient is the maximum load the process could ever remove. The constraint forces $e_{tr_i p} = 0$ when $\sigma_{tr_i} = 0$; when $\sigma_{tr_i} = 1$, it reduces to a valid upper bound on the maximum removable load and therefore does not further restrict the feasible removal.

At most $\overline{I}_{tp}\overline{V}_t$ of p can reach plant t, being the highest concentration it will accept multiplied by the maximum volume it can process daily, and the process can remove at most the fraction $H_{tr_i p}$ of that amount.

$$e_{tr_i p} \le H_{tr_i p}\, \overline{I}_{tp}\, \overline{V}_t\, \sigma_{tr_i}, \qquad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P},\ i = 1, \ldots, n_t. \tag{18}$$

A larger coefficient would still be valid, but it would unnecessarily expand the feasible region of the LP relaxation [2]. This is why, where a plant does not have a genuine inlet limit on p, $\overline{I}_{tp}$ is adjusted to the maximum value that the sources are capable of producing instead of being set to an arbitrarily large number.

Outlet water quality. The water leaving each plant must remain within the regulatory limits. For parameter p, $\ell_{tp}^{n_t}$ is the load remaining in the treated water, while $\sum_s b_{st}$ is the corresponding water volume, ensuring the volume-weighted average value of p leaving the plant is between $\underline{Q}_p$ and $\overline{Q}_p$.

$$\underline{Q}_p \sum_{s \in \mathcal{S}} b_{st} \le \ell_{tp}^{n_t} \le \overline{Q}_p \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P}. \tag{19}$$

The volume term is $\sum_s b_{st}$ rather than $\sum_z c_{tz}$; the two are equal by plant flow conservation, and utilising the inflow maintains consistency across all quality constraints expressed.

Because the current treatment model can only reduce parameter loads, treatment cannot correct a blend that is already below a required lower bound. Consequently, lower-bound compliance must be achieved through source selection and blending.

6 Water quality formulation logic

TODO: update the following section to reflect the new formulation, and add a discussion of how the transformed pH is handled - I think we should explicitly state that removal processes do not act on pH because this doesn't work - perhaps in linearity assumptions section?

Done :» I rephrased, added and copied some content :»

Although nicely generic water quality constraints are presented above, i.e. (13) to (19), we shall devote this section to interpret the quality parameters $p \in \mathcal{P}$ and how they are treated in plants. We confine ourselves to "removal process", though our model can be extended to include more treatment processes, e.g. "chemical dosing", and thus, this section will be adapted to be consistent with the existing parameter notation while preserving the linear structure of the MILP.

6.1 Inlet treatability and Outlet water quality

In (13), we propose acceptance criteria with respect to the incoming $p \in \mathcal{P}$ at every plant $t \in \mathcal{T}$. Due to linearity assumption, we insist these quality parameters can be blended approximately linearly by volume, and the source value can be used directly from Q_{sp}. In this constraint, we avoid division by decision variables b_{st}, which result in a non–linear program, we write the constraint in load form. It can be understood as a volume–weighted average quality, rather than pure quality constraints.

In (19), we restrict the outgoing $p \in \mathcal{P}$ after undergoing the treatment processes at $t \in \mathcal{T}$ to be within a certain regulatory range. Likewise, to avoid division by decision variables b_{st}, we express the constraints in load form, or equivalently, volume–weighted average.

6.2 Turbidity

Turbidity measures the cloudiness of water and is associated with suspended particles such as sediment, organic matter, algae, clay, or runoff material. In water supply modelling, turbidity is important because high turbidity may increase treatment requirements and affect disinfection performance [14, 8, 10].

For turbidity, the quality parameter is represented by setting $p = \text{turb}$. Thus, $Q_{s,\text{turb}}$ denotes the turbidity value of source s, while $\underline{Q}_{\text{turb}}$ and $\overline{Q}_{\text{turb}}$ denote the lower and upper turbidity limits.

Conveniently, the turbidity requirement is handled directly by replacing the sub–index p with turb. For example, (13) becomes:

$$\underline{I}_{t,\text{turb}} \sum_{s \in \mathcal{S}} b_{st} \leq \sum_{s \in \mathcal{S}} Q_{s,\text{turb}}\, b_{st} \leq \overline{I}_{t,\text{turb}} \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}.$$

6.3 Alkalinity

Alkalinity measures the ability of water to neutralise acid [9, 7, 12]. It represents the buffering capacity of water and is relevant to pH stability, corrosion control, treatment performance, remineralisation, and water stability in distribution systems [9, 12].

For alkalinity, the quality parameter is represented by setting $p = \text{alk}$. Thus, $Q_{s,\text{alk}}$ denotes the alkalinity value of source s, while $\underline{Q}_{\text{alk}}$ and $\overline{Q}_{\text{alk}}$ denote the lower and upper alkalinity limits.

Conveniently, the alkalinity requirement is handled directly by replacing the sub–index p with alk. For example, (13) becomes:

$$\underline{I}_{t,\text{alk}} \sum_{s \in \mathcal{S}} b_{st} \leq \sum_{s \in \mathcal{S}} Q_{s,\text{alk}}\, b_{st} \leq \overline{I}_{t,\text{alk}} \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}.$$

We highlight here that, according to [13], removal process can directly reduce alkalinity, though it can be increased by an adjustment process.

6.4 pH transformation

Unlike Turbidity and Alkalinity, pH cannot be treated in any removal processes, but rather in other treatment processes, e.g. chemical dosing, coagulation and lime softening [13].

More importantly, pH values are not averaged directly because direct pH averaging can produce a scientifically misleading blended value. Therefore, in our mathematical model, we consider its corresponding hydrogen ion.

$$\text{pH} = -\log_{10}([\text{H}^+]), \tag{20}$$

where $[\mathrm{H}^+]$ represents hydrogen ion concentration [1].

To keep the MILP linear, pH is handled as a preprocessing transformation [11]. For the pH quality parameter, $Q_{s,\mathrm{pH}}$ stores the transformed hydrogen ion concentration obtained from the raw source pH value:

$$Q_{s,\mathrm{pH}} = 10^{-\mathrm{pH}_s}. \tag{21}$$

The symbols $\underline{\mathrm{pH}}$ and $\overline{\mathrm{pH}}$ represent the raw regulatory lower and upper limits in pH units. These raw limits are converted once during data preparation before being passed into the MILP. If the acceptable raw pH range is:

$$\underline{\mathrm{pH}} \leq \mathrm{pH} \leq \overline{\mathrm{pH}}, \tag{22}$$

then the corresponding transformed bounds stored for the pH quality parameter are:

$$\underline{Q}_{\mathrm{pH}} = 10^{-\overline{\mathrm{pH}}}, \qquad \overline{Q}_{\mathrm{pH}} = 10^{-\underline{\mathrm{pH}}}. \tag{23}$$

The direction of the bounds changes because pH and hydrogen ion concentration are inversely related [5]. A higher pH corresponds to a lower hydrogen ion concentration, while a lower pH corresponds to a higher hydrogen ion concentration.

As mentioned earlier, pH cannot be handled by removal process, thereby, inapplicable to constraints from (15) to (19). Nonetheless, as an incoming quality parameter at $t \in T$, the constraint (13) is still valid. Formally, this is:

$$\underline{I}_{t,\mathrm{pH}} \sum_{s \in \mathcal{S}} b_{st} \leq \sum_{s \in \mathcal{S}} Q_{s,\mathrm{pH}}\, b_{st} \leq \overline{I}_{t,\mathrm{pH}} \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}.$$

6.5 Blended pH recovery after optimisation

The optimisation model constrains pH through the transformed value $Q_{s,\mathrm{pH}}$. After the solver finds the optimal source to plant flows, the blended hydrogen ion concentration at plant t is calculated as:

$$Q_{t,\mathrm{pH}}^{\mathrm{blend}} = \frac{\sum_{s \in \mathcal{S}} Q_{s,\mathrm{pH}} b_{st}}{\sum_{s \in \mathcal{S}} b_{st}}. \tag{24}$$

The blended pH is then recovered in post-processing as:

$$\mathrm{pH}_t^{\mathrm{blend}} = -\log_{10}\left(Q_{t,\mathrm{pH}}^{\mathrm{blend}}\right). \tag{25}$$

This recovered pH value is not an MILP decision variable. It is a post-solve reporting value used for validation, interpretation, dashboard display, and output generation.

TODO: update these to reflect new removal formulation i.e. last assumption is now incorrect and there should be more assumptions about the removal processes.

I think you meant the Section 6.4? Because I think the Section 6.5 is sound, and I modified few things in Section 6.4, also other sections before 6.4. One more thing, I added a separate subsection for 6.7 Assumptions for removal process.

6.6 Linearity assumptions for water quality

The water quality formulation relies on simplifying assumptions that preserve linearity in the MILP. The key assumptions are:

(i) Water from selected sources is completely mixed before quality is assessed.

(ii) The model represents a single time period, so temporal variation in water quality is not modelled.

(iii) Source quality values are fixed input parameters for the representative period.

(iv) Alkalinity and turbidity are approximated as volume-weighted linear blends.

(v) Raw pH values are not averaged directly; pH is transformed before optimisation.

(vi) No nonlinear chemical reactions between blended sources are modelled.

(vii) Water quality is checked at each treatment plant, using the blend of its incoming sources.

These assumptions allow the water quality constraints to remain linear and suitable for inclusion in the MILP formulation.

6.7 Assumptions for removal processes

In practices, several designs for removal processes exist, see [6, 15]. Therefore, we shall propose the following assumptions for convenience and simplicity.

- Removal process $\mathcal{R}_t$ is conducted in a sequence.
- Maximum removal efficiency H_{trp} is known. I'm thinking about a weaker assumption, not maximum but a fixed efficiency is known?
- Uniform treatment: $\sum_{s\in\mathcal{S}} b_{st}$ receives the same treatment.
- No water loss after passing through $r_i \in \mathcal{R}_t$.

7 Application to the toy model

TODO: toy model is now also incorrect because it doesn't account for removal processes.

The toy model is not a different problem but the special case of the network formulation with three sources, a single treatment plant and a single demand zone. With one plant and one zone there is nothing to route: the arc variables are fixed by conservation, and only the source decisions α_s and a_s remain free.

Every equation of the toy model follows from the general formulation by the substitutions below. Nothing is restated here.

Substitution	**Consequence**
$\mathcal{T} = \{t\}$, $\mathcal{Z} = \{z\}$	one plant, one zone
$\beta_t = 1$	the plant is always active, so F_t is a constant and drops out
$\gamma_{st} = \delta_{tz} = 1$	every arc is active
$b_{st} = a_s$	by source flow conservation
$c_{tz} = \sum_s a_s$	by plant flow conservation

The reduction is a substitution rather than a rewrite: any constraint indexed over $\mathcal{T}$ or $\mathcal{Z}$ contributes exactly one instance, and any variable on an arc is determined by conservation as a sum of source draws. Every term of the general model is inherited unchanged.

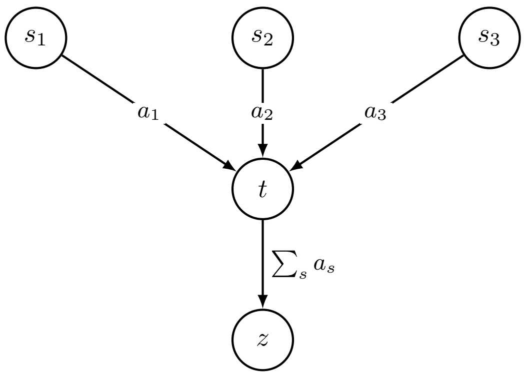

Figure 2: Toy model. Three sources draw a_1, a_2, a_3 into the single plant t, which delivers their sum to the single zone z.

Since this is just the network model with routing fixed, going the other way needs nothing new: free the routing variables and add more sources, plants and zones. The toy can therefore be built and tested first, then scaled up without reformulating.

References

[1] United States Environmental Protection Agency. pH. https://www.epa.gov/caddis/ph, 2026.

[2] Jeffrey D. Camm, Amitabh S. Raturi, and Shigeru Tsubakitani. Cutting big m down to size. *Interfaces*, 20(5):61–66, 1990. URL: http://www.jstor.org/stable/25061401.

[3] Castro, Teles, and Novais. Linear program-based algorithm for the optimal design of wastewater treatment systems. *Clean Technologies and Environmental Policy*, 11(1):83–93, 02 2009.

[4] Pedro M. Castro, Henrique A. Matos, and Augusto Q. Novais. An efficient heuristic procedure for the optimal design of wastewater treatment systems. *Resources, Conservation and Recycling*, 50(2):158–185, 2007.

[5] Fondriest Environmental. pH of Water - Environmental Measurement Systems. https://www.fondriest.com/environmental-measurements/parameters/water-quality/ph/, 2013.

[6] Lower Murray Water. Annual drinking water quality report 2022–2023. Technical report, Lower Murray Water, Victoria, Australia, 2023. URL: https://www.lmw.vic.gov.au/wp-content/uploads/2023/02/ADWQR-2022-2023-FA.pdf.

[7] Test Needs Pty Ltd. Understanding Water Alkalinity: Importance, Standards, and Testing in Australia. https://www.testneeds.com.au/understanding-water-alkalinity-importance-standards-and-testing-in-australia, 2025.

[8] T. Matos, M. S. Martins, R. Henriques, and L. M. Goncalves. A review of methods and instruments to monitor turbidity and suspended sediment concentration. *Journal of Water Process Engineering*, 64:105624, 2024. doi:10.1016/j.jwpe.2024.105624.

[9] NHMRC,NRMMC. *Australian Drinking Water Guidelines Paper 6 National Water Quality Management Strategy*. National Health and Medical Research Council and National Resource Management Ministerial Council and Commonwealth of Australia, Canberra, 2011. Version 4.0, Updated June 2025. URL: https://guidelines.nhmrc.gov.au/australian-drinking-water-guidelines.

[10] University of Minnesota. Metrics - Turbidity and Suspended Matter. https://water.rs.umn.edu/turbidity, 2017.

[11] World Health Organization. pH in Drinking-water. https://cdn.who.int/media/docs/default-source/wash-documents/wash-chemicals/ph.pdf, 2007.

[12] The Perfect Water. pH & Alkalinity in Drinking Water: Safe Levels Explained. https://www.theperfectwater.com/blog/post/a-guide-on-ph-alkalinity-and-their-impact-on-drinking-water-quality, 2025.

[13] World Health Organization, editor. *Water Treatment and Pathogen Control: Process Efficiency in Achieving Safe Drinking-Water*. World Health Organization and IWA Publishing, 2004. URL: https://www.who.int/publications/i/item/9241562552.

[14] World Health Organization. *Guidelines for Drinking-water Quality*. World Health Organization, Geneva, 4 edition, 2011. Fourth edition, incorporating first and second addenda. URL: https://iris.who.int/server/api/core/bitstreams/69c17edd-ee26-425b-9d34-33799377e886/content.

[15] Anastasios Zouboulis, George Traskas, and Petros Samaras. Improvement of drinking water plant treatment. *IASME Transactions*, 4(2):523–528, 2005.